

# Robot as a Peer in Learning

## Effect of Robot Role on Help Seeking and Comfort

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10-08-2026

### 1 Introduction

The idea for this project is pretty simple. When a student works with an educational robot, does the robot always have to act like a teacher? In most learning systems the robot is the one who knows the answers and corrects the student. That setup makes sense in a way, but it also puts the learner into a clear student-teacher relationship. I wanted to try something a little different, where the robot behaves more like another learner who happens to be working on the task with the student.

This question comes out of a bigger area of research on social robots in education. Belpaeme et al. looked at 101 studies and found that educational robots have positive effects on both learning and how students feel, but they also pointed out that most robots are built as tutors or teachers rather than peers [1]. Only a small part of the studies they reviewed used a robot acting as a peer or a novice. That is what made the peer role interesting to me, because a robot with less authority might change how a learner behaves even if the material itself stays exactly the same. If that turns out to be true it could matter outside a lab too, plenty of people avoid asking for help just because they don't want to look like they don't get something, and a robot that acts less like an authority figure might lower that barrier a bit, whether that is in a classroom or in some kind of self study app.

I decided to test this with a small football quiz. I did not pick football because I cared about testing football knowledge specifically. I picked it because some people know a lot about it and some know almost nothing, and also because a quiz feels more like a game than a formal exam. That mattered for the comfort part of the study, since I wanted participants thinking about answering and asking for help, not feeling like every wrong answer was being graded. In practice the system is a short browser based quiz, a robot avatar introduces itself out loud, then asks ten questions one at a time, and there is a help button on screen the whole time that behaves differently depending on which role the robot has been given.

Most of the similar studies I found either used children, a robot that was physically in the room, or a task spread across several sessions. My setup is different in a few ways at once, it uses adults, a robot that only ever appears on a screen, and one short quiz session, which let me test the same basic idea in a much smaller and cheaper format than a lot of the earlier work.

The main question was whether the robot’s social role changes two things that are easy to observe in a short interaction. First, help seeking, meaning how often someone pressed the “Ask for Help” button. Second, comfort, meaning how comfortable a participant said they felt once the task was over. I also looked at the final quiz score, mainly to check whether the social framing changed how well people did.

My two expectations were simple, so I will just list them here to make them easy to find:

- **H1:** Participants in the peer condition would ask for help more often than participants in the tutor condition.
- **H2:** Participants in the peer condition would report higher comfort than participants in the tutor condition.
- **H3:** There would not be a large difference in quiz score between the two conditions, since the study was mostly about willingness to interact and not about making the quiz easier for one group.

While I was working on this, I realised the three measures are not as separate as they first look. If the help button makes the quiz a lot easier, then asking for help can directly raise the score. And a higher score can then make someone feel more comfortable afterwards. So a result that looks social could actually be coming from a small detail in how the system was built, and this became an important part of how I ended up reading the final data.

So this study is not presented here as a clean demonstration that peer robots beat tutor robots. I see it more as a small, practical test. It let me try out the interface, the role manipulation, the logging, and the main measures, and it also showed me where the design still needs work, which in my view is a useful outcome for a small user study.

## 2 Related Work

Several earlier studies helped me decide what to change in the robot and what to measure, tracing a slow shift in educational robotics from treating the robot mainly as a tutor toward considering it as a possible peer. I did not try to reproduce any of these studies exactly. Instead I used them as guidance for the basic idea and then adapted the interaction to fit a small browser based experiment.

Belpaeme et al. give the broad background for this project. Their review describes social robots as useful educational partners and reports benefits for both learning and affective outcomes [1]. At the same time they note that most educational robot studies use a teacher or tutor role, and the peer or novice role is much less common. This matters because changing a robot’s social role can change how a learner understands the whole interaction. A learner might feel evaluated by a tutor but more relaxed with a partner.

Zaga et al. looked at a peer versus tutor distinction in a guided learning task with children [2]. Their study was especially useful for my design because the social character of the robot came from its behaviour and language, not from different hardware. I followed the same logic. The robot in my experiment looked the same in both conditions, while

the wording and the support style changed, which makes the comparison easier to read since there is no second physical robot that could explain the result.

Kanda et al. also found that a peer styled robot can influence attention and how children respond [3]. This gave me the idea that the peer role could be more than just one friendly sentence on a screen, and that the way a robot talks and shares the task can add to the feeling of working together. My study is much smaller and runs in a browser, so I do not treat it as a direct replication.

A particularly relevant study for an adult sample is Salomons et al. [4]. They compared peer and tutor robots in an electronic circuit learning task and did not find one simple overall advantage across all outcomes. Instead the peer condition helped adults with lower prior knowledge, and participants rated the peer robot positively on several social measures. This told me that peer effects can depend on the person and the task, rather than producing one big improvement for everybody.

Diyas et al. compared peer and teacher robot roles in a programming task [5]. This was another example of changing the social framing while keeping the learning content comparable, which supported my choice to use the same question pool in both conditions and put the manipulation into the robot’s role and help behaviour instead of building two different tasks.

The protégé effect described by Chase et al. is related to the idea of a study buddy [6]. Their work suggests that how a learner is positioned next to another agent can change how much effort they put in. My system never asked participants to teach the robot, so it is not a direct test of the protégé effect, I only borrowed the general idea that a shared task can feel different from a one sided teaching interaction.

Dillenbourg’s writing on collaborative learning was useful when I was thinking about language [7]. Collaboration is not only about two agents being in the same place, the roles and goals of the people involved matter too. For that reason the peer robot used words like “we” and “let’s” in its help messages, since I wanted the interaction to sound cooperative rather than just polite.

Bainbridge et al. compared physically present robots with agents shown on video [8]. Their findings connect to one of the biggest limitations of my study, because my robot only ever appeared on a laptop screen and was never physically in the room. Whatever social feeling a real robot body creates could be stronger, weaker, or just different from what I measured here.

Taken together, these studies pointed me in a fairly clear direction: two social roles, one general task, a measure of help seeking, and a measure of comfort. At the same time the literature also made me cautious about drawing big conclusions from a small sample. The final design followed the basic peer versus tutor idea, but the results still have to be read alongside the exact mechanics of the help function.

### 3 System Description

I built the experiment as a small Flask web application. The participant opens a browser page and sees a robot themed interface. The main screen has the robot image, a short message, the quiz area, the answer options, and two main actions: submitting an answer and asking for help. I kept the interface simple on purpose, since the focus of the study was the interaction and not the visual design.

### 3.1 System Architecture

On the backend this is one Flask application written in Python, with the frontend built from basic HTML and CSS plus a little bit of JavaScript for things like the speech synthesis calls and the help button. When someone opens the link, the server starts a session and, inside that session, randomly picks either the peer or the tutor condition, so the condition lives in the session data rather than anything the browser decides on its own. The server also keeps track of which question set was chosen, which question the participant is currently on, and how many help requests have been used so far.

Each screen is rendered from a template, and the same HTML structure gets reused for both conditions, only the wording and a few CSS classes controlling tone change depending on the session's condition value. This is basically why the two conditions look almost identical, the real difference is in what text gets inserted into the same layout, not in the layout itself.

Every important action, submitting an answer, pressing help, moving on to the next question, gets written to a log as soon as it happens, along with a participant ID, the condition, the question number, and a timestamp. That log ended up being the backbone of the whole system, it is what let me go back after the study and look closely at things like Participant 2007's pattern of behaviour instead of only trusting the final score.

### 3.2 Interface and Role Assignment

The server assigns the condition before the main page loads. A participant is placed randomly into either the peer or the tutor condition, they do not choose the role themselves. The page then states which type of companion has been assigned. In the peer condition the robot is described as a study buddy, in the tutor condition it is described as an instructor. Figure 1 shows the first screen, which introduces this role assignment. The visual design is almost identical across the two conditions on purpose, since I wanted participants to notice the role through the message and the interaction style rather than through a completely different looking page.

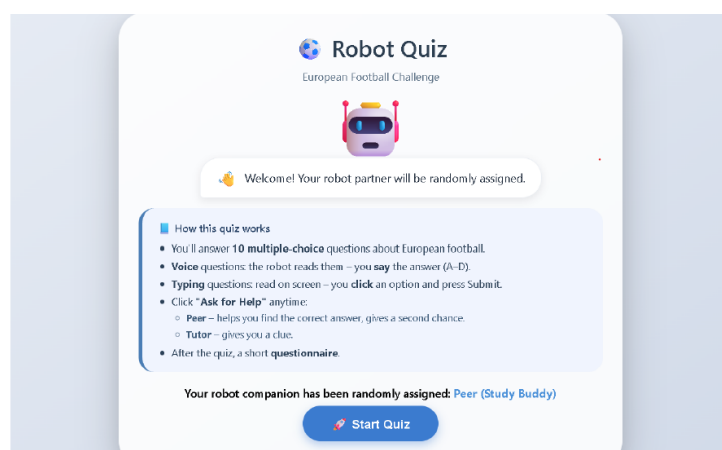


Figure 1: Opening screen of the robot quiz. The page introduces the robot and informs the participant about the assigned role.

After the participant clicks start, the robot gives a spoken introduction using the

browser’s speech synthesis. The quiz does not begin until that message finishes. The wording is more relaxed in the peer condition and more instructional in the tutor condition. The peer message talks about solving the questions together and mentions that another chance is available after a wrong answer. The tutor message is more formal and asks the participant to think carefully.

This opening is part of the social manipulation and not just an instruction screen. A participant hears the robot’s attitude before they even see the first question, and I think that matters, because the first few seconds of an interaction can set expectations about whether a person is going to be judged or supported.

### 3.3 Quiz Structure

The quiz has ten multiple choice questions. Questions come from two sets, A and B, and one set is picked at random. The questions are about European football and mix familiar facts with less familiar ones, so the quiz would not only test one narrow level of football knowledge.

Figure 2 shows the main quiz screen. The participant sees a question and four answer choices, and depending on the question the interface can accept a spoken response or a selected answer. There is also an “Ask for Help” button, and every important action gets logged so that later I could look not only at the final score but also the sequence of help requests.

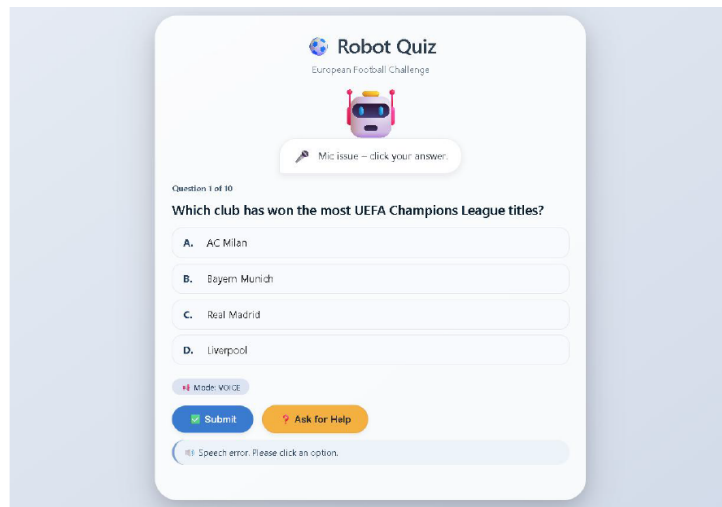


Figure 2: Example quiz screen with four answer options and the help button.

### 3.4 Help Mechanism

The biggest difference between the two conditions is the help mechanism. In the peer condition, asking for help removes one wrong option from the screen, and the robot uses friendly language when it suggests leaving one option out. The participant can ask again and another wrong answer gets removed, so after three useful requests in principle only the correct option is left.

In the tutor condition, the robot returns a short factual clue related to the question. The answer choices stay on the screen, and the participant still has to connect the clue to one of the four options, which makes the tutor hint quite a bit less direct than the peer hint.

At the time I built this, the peer condition also allowed one extra attempt after a wrong first answer, and the tutor condition did not have that extra attempt. At first this felt like just another way of making the study buddy more forgiving, but by the time I looked at the final analysis it was clear this was also a difference in task difficulty.

Figure 3 shows the peer side of the interaction before the quiz starts. I include it because the message helps explain what the participant expects from the robot going in, and the friendly wording together with the promise of another attempt may itself change how safe it feels to get something wrong.

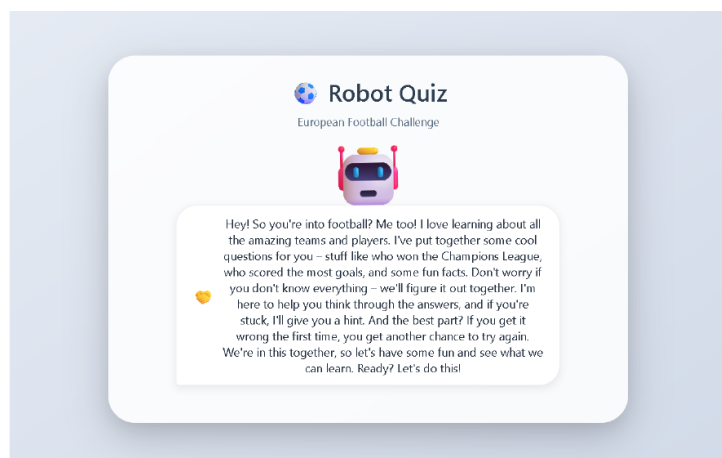


Figure 3: Example of the peer robot explanation before the quiz begins.

### 3.5 Early Attempt with a Language Model

I also tried a dynamic language model version of the peer hints, with the idea being more varied and natural responses. In practice the local model was too slow for a short quiz interaction and sometimes gave unreliable information, so I stopped using it and switched to fixed templates instead. The templates were faster, factually controlled, and much easier to log.

This turned out to help in another way too. The help log could tell me exactly which kind of hint was used and how often a participant asked for it, and that detailed record is what later let me notice that at least one participant was using the help button as a systematic way to solve answers. Without the log I probably would have just seen a high final score and assumed it reflected normal performance.

## 4 Methods

### 4.1 Participants

I ran the study with fourteen participants, seven in the peer condition and seven in the tutor condition. The participants were university affiliated adults aged between 23 and 26. Twelve were male and two were female. Recruitment happened through personal and university contacts rather than any formal sign up system, so this is a convenience sample and not one drawn randomly from a wider population. The sample is small, so I am not using it to make any claims about demographic effects, it was mainly meant for a first practical test of the system.

### 4.2 Design

The experiment used a between subjects design, each person experienced only one robot role. I chose this because giving the same person both conditions could create a carry over effect, a participant who first works with a very friendly study buddy might respond differently to a formal tutor later, even with the same quiz underneath.

The role was assigned randomly by the server. There was no button on the page for the participant to pick their role, the system chose the condition before the page was even rendered, which keeps the assignment separate from any participant preference.

The two conditions used the same question structure and the same general interface. A random choice between question sets A and B was made when the session started, which was meant to lower the chance that one group would only get an unusually easy or unusually hard set.

### 4.3 Procedure

The session followed a simple sequence. First the participant read the study information and, where the ethics procedure applied, gave consent. The web application then assigned a condition. The robot gave its condition specific opening message. The participant completed the ten questions, and during the quiz the system logged answers and help requests. Finally the participant filled in a short questionnaire and was thanked.

The whole session was planned to take around twenty minutes. The experiment ran remotely, and participants accessed the quiz through the web interface. I kept the setup simple on purpose so the same interaction could run without needing a physical robot.

The post task questionnaire included a comfort rating on a 1 to 7 scale. A manipulation check asked whether the robot behaved more like a classmate or more like a teacher. Age and gender were also collected as basic demographic information.

### 4.4 Measures and Analysis

I used three main measurements. Help seeking was the number of times a participant pressed “Ask for Help” during the quiz. Comfort was the self reported rating after the task. Task score was the number of correct answers out of ten. Neither the comfort item nor the manipulation check question came from an existing published scale, I wrote both of them myself for this short session, mostly because the standard comfort or engagement

scales I came across are built for longer surveys and would not really have fit a twenty minute study. The first two were the main outcomes tied to the research questions, the score was more of a control measure, since I did not want a possible social effect to get confused with a simple difference in correct answers.

The analysis used independent samples  $t$  tests. The help count and comfort comparisons followed the directional expectations, while the score comparison was treated as two sided. I also calculated Cohen's  $d$  to describe how large the differences were. The significance level was set to  $\alpha = .05$ .

One extra part of the method turned out to matter later. Alongside the statistical tests, I also treated the interaction logs as a qualitative source, going through individual sequences of help requests case by case instead of only looking at the final help count. This is because a help request can mean different things, someone might ask once because they are confused, or press the button several times because the system gives them an efficient way to solve the question. Counting both of those the same way is not really fair, so the raw log gave a more realistic picture of what was actually happening.

The manipulation check was reassuring here. Every participant in the peer condition described the robot as a classmate, and every participant in the tutor condition described it as a teacher. None of the fourteen participants had to be removed, which suggests the role wording was noticed as intended, although it does not prove that role perception was the only difference between the conditions.

## 5 Results

### 5.1 Help Seeking

The first thing I noticed was that the peer condition had a higher average number of help requests. Participants in the peer condition made an average of 12.14 requests, with a standard deviation of 11.84. In the tutor condition the average was 6.00 requests, with a standard deviation of 3.21, so the peer group asked for help about twice as often.

That difference sounds large, but the statistical test did not cross the usual significance threshold. The one tailed independent samples  $t$  test gave  $t(12) = 1.325$  and  $p = .105$ . The effect size was  $d = 0.71$ , a medium to large difference, but the sample was too small and the data too variable to treat this as convincing evidence for H1.

Part of the problem is how spread out the data is. In the peer group, two participants did not ask for help at all, while one made 30 requests. The tutor group's largest help count was much lower than that. So the peer average is being pulled strongly by one unusual pattern, and with only seven people per condition there is very little protection against that kind of effect.

### 5.2 Comfort

The comfort result was much clearer. The peer group reported an average comfort score of 5.71 out of 7, compared with 3.57 in the tutor group. The standard deviations were 0.95 and 0.79. The one tailed test gave  $t(12) = 4.593$  with  $p < .001$ , and Cohen's  $d$  came out to 2.46, which on the surface is a very strong difference.

This is probably the result that looks most encouraging for the peer idea at first glance. Participants with the study buddy generally reported a more comfortable experience, and the gap is not small. Even so, I do not think it is fair to say the robot’s peer role alone caused this, because the task itself was easier in the peer condition.

### 5.3 Score

The final score also showed a large difference. Peer participants answered an average of 7.29 questions correctly out of ten, with a standard deviation of 2.36. Tutor participants averaged 3.57, with a standard deviation of 1.72. The two tailed test gave  $t(12) = 3.366$  and  $p = .006$ , with  $d = 1.80$ .

This was not what I expected. I wanted the two conditions to score similarly, so that a comfort difference would not simply mean one group got more answers right, but instead the peer group scored almost twice as high.

Table 1 puts all three measures next to each other so the overall pattern is easier to see in one place.

Table 1: Summary statistics for help requests, comfort, and score across the two conditions.

| Measure           | Peer M (SD)   | Tutor M (SD) | $t(12)$ | $p$   | $d$  |
|-------------------|---------------|--------------|---------|-------|------|
| Help requests     | 12.14 (11.84) | 6.00 (3.21)  | 1.325   | .105  | 0.71 |
| Comfort (1–7)     | 5.71 (0.95)   | 3.57 (0.79)  | 4.593   | <.001 | 2.46 |
| Score (out of 10) | 7.29 (2.36)   | 3.57 (1.72)  | 3.366   | .006  | 1.80 |

Figure 4 gives a visual summary of the three main outcomes.

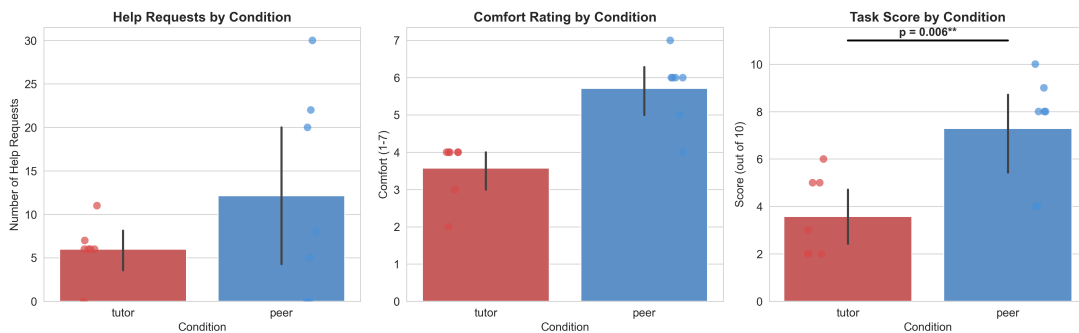


Figure 4: Mean help requests, comfort ratings, and quiz scores for the peer and tutor conditions.

I think the most useful way to read the score result is to look back at how help worked. In the peer condition every help request could remove one wrong answer, which is a powerful kind of assistance, a participant could keep asking and gradually cut four options down to one. The tutor clue never worked that way, it still required the participant to reason through the question.

## 5.4 A Closer Look at the Help Log

One participant made this difference especially obvious. Participant 2007 requested help three times on almost every question. The logged sequence shows three options being eliminated one after another, and this participant ended up scoring 10 out of 10 while making 30 help requests, the highest help count and the highest score in the whole dataset.

I do not think it is fair to call all thirty of those requests normal help seeking. This person found a very effective strategy that the interface itself made possible, they were not really asking for reassurance or explanation, the help button had basically become part of an answer elimination method.

This changes how I read H1 too. The average help count was higher in the peer condition, but that number mixes at least two behaviours together, asking for support because you are unsure, and asking repeatedly because the system will remove an option for you. Those are not the same psychological behaviour, even though they get counted the same way in a simple average.

## 5.5 Manipulation Check

All seven peer participants described the robot as a classmate, and all seven tutor participants described it as a teacher. So the basic role distinction was understood, and I never had a case where someone in the peer condition thought the robot was acting like a teacher, or the reverse.

Overall the results are mixed rather than simply positive. Help seeking was higher in the peer condition but not statistically significant. Comfort was clearly higher for the peer group. Score was also clearly higher, but that was not predicted and is hard to separate from the stronger help mechanism.

# 6 Discussion

The main question of the study was whether turning a robot from a tutor into a peer changes how people ask for help and how comfortable they feel during a learning task. Looking at the results and the interaction logs together, my answer is partly yes, but I would not call this a clean peer effect.

## 6.1 The Comfort Result

The comfort result is the strongest part of the study. The peer condition got a much higher comfort rating, and the effect size was very large. This fits the general idea in the literature that a less authoritative social role can make an educational interaction feel more positive [1, 4], and it also matches how the peer robot was written, since the peer message used casual phrases, talked about solving the quiz together, and tried to make mistakes feel less serious.

At the same time the peer condition had two technical advantages the tutor condition did not have. First was the option elimination help, second was the extra attempt after a wrong answer. Both of these can make the task easier, and since peer participants also

scored much higher, I cannot fully separate comfort caused by social framing from comfort caused by simply doing better.

This is an important lesson from the study for me. My original plan treated the quiz score as something that should stay roughly similar across conditions, which would have let me say that any difference in comfort or help seeking was probably social. But the implementation itself changed how hard the task was, so the result ended up being partly a test of the help interface rather than only a test of robot role.

## 6.2 The Help Seeking Result

The help seeking result needs the same kind of caution. The peer average was about twice the tutor average and the effect size was not small, but  $p = .105$  means I cannot reject the null hypothesis at the threshold I set. More importantly, one participant accounts for thirty of the peer requests, and once a system makes repeated help requests useful for actually solving the question, a high frequency does not necessarily mean a high willingness to ask a human-like partner for help.

I think this is one of the clearest arguments for checking interaction logs and not just summary statistics. Looking only at the average could suggest a simple story, that participants felt more willing to ask the peer robot for help. The log tells a more complicated story, that the interface taught at least one participant a very effective way to use the help button.

The manipulation check is more reassuring on its own. All participants understood the intended role, so the wording was not lost on them, they generally knew whether they were dealing with a classmate-like robot or a teacher-like one.

## 6.3 Limitations

The robot was never physically embodied, the participant only ever saw a browser page with a robot image and speech. Bainbridge et al. reported that physically present robots can produce stronger effects than agents shown through video [8], and Belpaeme et al. also discuss embodiment as an important part of educational robot interaction [1]. So my results cannot be assumed to carry over to a robot actually standing next to a student in a classroom.

The small sample is another limitation. Fourteen participants is enough to find problems in an interface, but not enough for a confident general conclusion. The help count especially was noisy, and a bigger sample would give a more stable estimate, although a bigger sample alone would not fix the unequal help mechanisms.

The demographic balance was limited too. The group was mostly male, and everyone fell into a narrow adult age range. That is not a problem for understanding this particular pilot, but it does limit how far the result can be generalised.

## 6.4 What I Would Change

If I ran this again, I would first make the two help systems informationally equal. Both conditions could get exactly one type of hint carrying the same amount of information. I would remove the extra retry from whichever condition had it and give both groups

the same answer policy. The opening spoken explanations should also match in length and information, the peer message should not reveal a special safety net that the tutor message does not mention.

I would also put a limit on repeated help requests, or at least record how many unique questions a person asked for help on. That would make it easier to tell “I asked because I was unsure” apart from “I asked until the system basically solved it for me.” Another useful measure would be the time between a question appearing and the first help request, which could show whether participants leaned on help immediately or only after trying the question on their own first.

Salomons et al. is a useful comparison here, since their adult study did not find a simple overall learning advantage for the peer robot [4]. So the fact that my peer group scored much higher is not something I would treat as evidence that peer robots directly improve learning, the mechanical difference in the hint system is a much simpler explanation for that result.

## 7 Conclusion

Even with these limitations, I think the project was useful. The role manipulation worked, the interface collected detailed logs, and the comfort measure showed a strong difference. More importantly, the study exposed a design mistake that would have been easy to miss. A social robot can be made friendlier and more supportive, but the support has to be balanced with just as much care as the robot’s language.

My overall conclusion is cautious. The peer robot seems to create a more comfortable interaction in this small experiment, and the help seeking pattern also points in the expected direction. But the study does not isolate the social role well enough for me to claim that the peer identity itself caused these differences, since the stronger help mechanism and the retry rule were changing the task at the same time.

For me the main result is not that one condition simply won. The more useful result is seeing how quickly a small interface detail can change what a behavioural measure actually means. When the goal is to study help seeking, the help tool itself needs to be built so that using it measures help seeking, not a shortcut to the answer.

This also points to a practical direction for future educational robot studies. The robot does not need to become less helpful to be studied fairly, the amount of information its support gives just needs to be controlled carefully. Friendly language, collaborative wording, and a non-authoritative role can then be tested without also changing how easy the task is at the same time.

So I see this study as a first step rather than a final answer. It gave me evidence that the peer framing is worth studying further, but it also showed that clean experimental design matters just as much as how the robot looks or how it talks.

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GitHub Link- <https://github.com/LikhithKumar555/Robot-as-a-Peer-in-Learning.git>